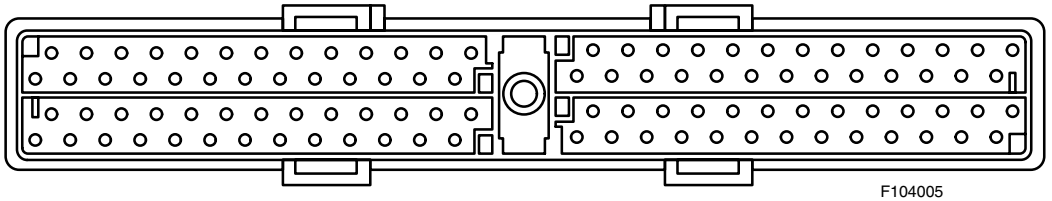
C175 (GY)

12A581

Powertrain Control Module (PCM) (12A650)



* 5.4L supercharged
** 4.2L
*** 4.6L, 5.4L Supercrew
† 5.4L Bi–Fuel
†† Natural Gas Vehicle (NGV)
††† except 4.2L
‡ 5.4L CNG
‡‡ 5.4L
‡‡‡ MSOF
◆ 4.6L, 5.4L
◆◆ except 5.4L Natural Gas Vehicle (NGV)
◆◆◆ except 5.4L supercharged
⁂ manual transmission
⁂⁂ automatic transmission

Pin	Circuit	Circuit function
1	**315 (VT/OG)	Shift solenoid B, control
	1025 (OG/YE)	Coil On Plug (COP) 6 (12029), control
2	–	not used
3	**1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	570 (BK/WH)	Ground
4	‡‡‡1496 (PK)	Transfer case speed sensor, signal
5	–	not used
6	◆237 (OG/YE)	Shift solenoid A, control
7	–	not used
8	**190 (WH/OG)	Intake Manifold Runner Control (IMRC) module, monitor
9	–	not used
10	–	not used
11	◆315 (VT/OG)	Shift solenoid B, control
12	◆911 (WH/LG)	Transmission Control Indicator Lamp (TCIL), control
13	107 (VT)	Module programming signal
14	784 (LB/BK)	Four–wheel drive, indicator, Low, signal
15	915 (PK/LB)	SCP Bus –
16	914 (TN/OG)	SCP Bus +
17	–	not used
18	–	not used
19	27 (OG/LG)	Fuel pump high/low relay, control
20	‡‡924 (BN/OG)	to transmission coast clutch solenoid, control
	***924 (BN/OG)	not used, ends in harness
21	349 (DB)	Crankshaft position sensor (6C315) –
22	350 (GY)	Crankshaft position sensor (6C315) +
23	–	not used
24	**570 (BK/WH)	Ground
25	567 (LB/YE)	Ground
	†48	Shield

Pin	Circuit	Circuit function
26	**526 (DB/LG)	coil driver (CD) – A control
	1024 (LG/WH)	Coil On Plug (COP) 1 (12029), control
27	**237 (OG/YE)	Shift solenoid A, control
	1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
28	–	not used
29	224 (TN/WH)	Transmission control switch, output, signal
30	–	not used
31	–	not used
32	◆◆311 (DG/VT)	Knock sensor (12A699) –
33	–	not used
34	◆1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
35	◆◆392 (RD/LG)	Heated Oxygen Sensor (HO2S) #12 (9G444), signal
36	968 (TN/LB)	Mass Air Flow (MAF) sensor (12B579), signal, return
37	923 (OG/BK)	Transmission Fluid Temperature (TFT) sensor, input
38	*1291 (RD/YE)	Intake air temperature sensor, 2 signal
39	743 (GY)	Intake Air Temperature (IAT) sensor (12A697), input
40	238 (DG/YE)	Fuel pump relay, switched power, output
	*786 (RD)	Crash fuel shutoff relay, output
41	347 (BK/YE)	A/C pressure switch, signal
42	**367 (BN)	Intake Manifold Runner Control (IMRC) module, signal
	*965 (LG/VT)	Supercharger bypass solenoid, control
43	*369 (WH/OG)	Charge air cooler pump relay, control
44	–	not used
45	–	not used

Pin	Circuit	Circuit function
46	†††367 (BN)	Intake Manifold Tuning Valve (IMTV), control
	◆◆367 (BN)	Intake Manifold Tuning Valve (IMTV), control
	‡‡367 (BN)	not used, ends in harness
47	360 (BN/PK)	EGR Vacuum Regulator (EVR), control
48	–	not used
49	1145 (LB/BK)	Digital Transmission Range (DTR) sensor, TR2
50	1143 (WH/BK)	Digital Transmission Range (DTR) sensor, TR4
51	570 (BK/WH)	Ground
52	**527 (RD/LB)	coil driver (CD) – C control
	1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
53	†††1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
54	480 (VT/YE)	Torque Converter Clutch (TCC) solenoid, control
55	729 (RD/WH)	Voltage supplied at all times (overload protected)
56	◆◆191 (LG/BK)	EVAP canister purge valve, control
57	◆◆310 (YE/RD)	Knock sensor (12A699)
58	–	not used
59	**1496 (PK)	Transfer case speed sensor, signal
	970 (DG/WH)	Turbine Shaft Speed (TSS) sensor, signal
60	74 (GY/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472), signal
61	393 (VT/LG)	Heated Oxygen Sensor (HO2S) #22 (9G444), signal
62	791 (RD/PK)	Fuel tank pressure transducer sensor (9C968)
	††1164 (LB)	Fuel rail temperature sensor, input
63	*356 (DB/LG)	Barometric Absolute Pressure (BAP) sensor, monitor
	††141 (RD/PK)	Injector pressure sensor, signal
64	⁂⁂199 (LB/YE)	Digital Transmission Range (DTR) sensor (7F293), TR3A
	⁂199 (LB/YE)	Voltage supplied in Start (overload protected)
65	352 (BN/LG)	Differential Pressure Feedback EGR (DPFE) sensor, signal
	††48	not used, ends in harness
66	1102 (YE/LG)	Cylinder–head temperature sensor (6G004), signal
67	◆◆91 (VT/WH)	Canister vent, control
68	679 (GY/BK)	vehicle speed signal
69	331 (PK/YE)	A/C clutch relay, control
70	1017 (BK/WH)	Check cap indicator lamp, control

Pin	Circuit	Circuit function
71	361 (RD)	Voltage supplied in Start and Run (overload protected)
72	561 (TN/RD)	Fuel injector 7, control
	‡1171 (WH)	Fuel injector 7, control
	††1171 (WH)	Fuel injector 7, control
73	559 (TN/BK)	Fuel injector 5, control
	‡1169 (WH/LB)	Fuel injector 5, control
	††1169 (WH/LB)	Fuel injector 5, control
74	557 (BN/YE)	Fuel injector 3, control
	‡1167 (WH/BK)	Fuel injector 3, control
	††1167 (WH/BK)	Fuel injector 3, control
75	555 (TN)	Fuel injector 1, control
	‡1165 (WH/RD)	Fuel injector 1, control
	††1165 (WH/RD)	Fuel injector 1, control
76	◆◆◆570 (BK/WH)	Ground
77	*570 (BK/WH)	Ground
	**570 (BK/WH)	Ground
78	**528 (PK/WH)	coil driver (CD) – B control
	1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
79	**911 (WH/LG)	Transmission Control Indicator Lamp (TCIL), control
	1029 (WH/RD)	Coil On Plug (COP) 8 (12029), control
80	926 (LB/OG)	Fuel pump relay, control
81	925 (WH/YE)	Electronic Pressure Control (EPC) solenoid, control
82	–	not used
83	264 (WH/LB)	Idle Air Control (IAC) valve (9F715), control
84	136 (DB/YE)	Output Shaft Speed (OSS) sensor (7M101), signal
85	795 (DG)	Camshaft position sensor (6B288), signal
86	–	not used
87	94 (RD/BK)	Heated Oxygen Sensor (HO2S) #21 (9F472), signal
88	967 (LB/RD)	Mass Air Flow (MAF) sensor (12B579), signal
89	355 (GY/WH)	Throttle Position Sensor (TPS) (9B989), signal
90	351 (BN/WH)	Reference, voltage
91	359 (GY/RD)	signal, return
92	810 (RD/LG)	Brake pedal position switch (13480), output
93	387 (RD/WH)	Heated Oxygen Sensor (HO2S) #21 (9F472), heater, control
94	388 (YE/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472), heater, control

Pin	Circuit	Circuit function
95	◆◆389 (WH/BK)	Heated Oxygen Sensor (HO2S) #12 (9G444), heater, control
96	◆◆390 (TN/YE)	Heated Oxygen Sensor (HO2S) #22 (9G444), heater, control
97	361 (RD)	Voltage supplied in Start and Run (overload protected)
98	562 (LB)	Fuel injector 8, control
	‡1172 (YE)	Fuel injector 8, control
	††1172 (YE)	Fuel injector 8, control
99	560 (LG/OG)	Fuel injector 6, control
	‡1170 (YE/LB)	Fuel injector 6, control
	††1170 (YE/LB)	Fuel injector 6, control
100	558 (BN/LB)	Fuel injector 4, control
	‡1168 (YE/BK)	Fuel injector 4, control
	††1168 (YE/BK)	Fuel injector 4, control
101	556 (WH)	Fuel injector 2, control
	‡1166 (YE/RD)	Fuel injector 2, control
	††1166 (YE/RD)	Fuel injector 2, control
102	–	not used
103	570 (BK/WH)	Ground
104	†††1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control